

**Systems Engineering Foundation for
U.S. Leadership in Agentic AI:
Georgia Tech Response to the AI Action Plan RFI**

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In response to:

Request for Information on the Development of an Artificial Intelligence Action Plan

Submitted by:



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1. Executive Summary

The Georgia Institute of Technology welcomes the opportunity to contribute to the development of an AI Action Plan that fulfills the vision of Executive Order 14179 to “sustain and enhance America’s global AI dominance.” As a global leader in AI research, education, and innovation—ranked 5th nationally in AI—Georgia Tech offers a comprehensive systems engineering perspective that directly addresses the policy objectives outlined in the Executive Order.

Our response frames AI leadership—with particular emphasis on agentic AI systems—through a systems engineering lens that naturally removes barriers to innovation while accelerating market-driven development of AI technologies. This systems-framed approach stands in contrast to component-focused strategies that often fail to realize the full potential of AI capabilities when integrated into real-world applications and systems. As agentic AI capabilities expand to perform increasingly autonomous tasks and make complex decisions, systems thinking will become even more critical for ensuring reliability, robustness, and alignment with human objectives.

Drawing on our institution’s unique position at the intersection of cutting-edge research and practical implementation—through our academic programs, our commercial research focus through the AI-Hub, and our applied research arm, the Georgia Tech Research Institute (GTRI)—we provide strategic principles and concrete policy recommendations to fulfill the objectives laid out in Executive Order 14179. Our recommendations establish frameworks that enable rapid, reliable, and robust advancement of American AI capabilities while preserving market dynamics and removing unnecessary impediments to innovation.

Key contributions to the 180-day AI Action Plan include:

- A systems engineering framework that accelerates AI innovation (“Engineering for AI”) while ensuring reliability, robustness, and performance
- Mechanisms for implementing AI to transform traditional engineering disciplines (“AI for Engineering”)
- Strategies for developing both “Workforce-ready AI” and an “AI-ready Workforce”
- Recommendations for computing infrastructure that enables next-generation AI capabilities
- Frameworks for objective evaluation and performance assessment of AI systems
- AI lifecycle management approaches that create sustainable competitive advantages

1.1. Georgia Tech: At the Intersection of AI Research and Implementation

The Georgia Institute of Technology brings unique qualifications to the development of an AI Action Plan that will advance American leadership:

- **National AI Leadership:** Ranked 5th nationally in AI, Georgia Tech graduates more AI talent than any other institution in the United States, creating a robust pipeline for industry and government.
- **Dual Research Focus:** Through our academic departments and the Georgia Tech Research Institute (GTRI), we maintain parallel excellence in both foundational AI research and applied national security implementations.
- **Tech AI Initiative:** Our specialized AI Hub (Tech AI) focuses on responsible AI for societal impact, aligning technical advancement with practical applications across multiple domains, with a strong focus on AI for engineering (e.g., supply chains, manufacturing, health care, agriculture)
- **Industry Integration:** Deep partnerships with leading technology companies, as well as leading engineering companies, facilitate rapid translation of research innovations into market-ready applications.
- **Interdisciplinary Strengths:** Georgia Tech combines significant strengths in AI technologies with domain expertise across numerous sectors of engineering and science, enabling systems-level integration.

This unique positioning allows Georgia Tech to bridge the gap between theoretical advancement and practical implementation—precisely the capability needed to develop an AI Action Plan that will accelerate American AI leadership while ensuring reliable, robust systems that can be deployed at scale.

2. Introduction: Systems Engineering as the Foundation for AI Leadership

Executive Order 14179 establishes a clear mandate to “sustain and enhance America’s global AI dominance” by removing barriers to innovation while promoting economic competitiveness and national security. Achieving this objective requires a paradigm shift in how we conceptualize and implement AI advancement—moving from isolated component optimization to integrated systems engineering, particularly as AI systems evolve toward increasingly agentic capabilities.

The historical evolution of AI reveals a critical inflection point in modern approaches. Early successes like IBM’s Deep Blue demonstrated remarkable capabilities in narrow domains but lacked the flexibility and adaptability required for broader applications. Today’s leading AI organizations — from DeepMind’s groundbreaking Alpha-Series innovations to DeepSeek’s demonstration of exceptionally engineered foundation model implementations — have embraced systems-level integration that orchestrates multiple AI components into cohesive, powerful solutions. Moreover, many of the fundamental advances in science and engineering now combine data-driven and model-drive approaches,

This transformation is especially evident in the rise of agentic AI—systems capable of autonomous goal-setting, planning, and execution across diverse tasks. By 2028, AI agents

are projected to be embedded in 33% of enterprise applications, up from less than 1% in 2024, with at least 15% of daily business decisions expected to be made autonomously by agentic AI. This shift from assistive to autonomous AI systems represents both the greatest opportunity and challenge for American technological leadership.

The concept of Agentic AI envisioned here goes well beyond the orchestration of large language models that combine natural language understanding and tool use; this field will develop to include planning systems and hierarchical memory and reasoning mechanisms, both parametric and symbolic, as well as goal-setting, extrospective perception, and multifaceted introspective and allocentric theories-of-mind capable of navigating human-centric dynamics in high-consequence applications. This expansive vision of agentic AI—with advanced cognitive architectures that can reason across multiple levels of abstraction and adapt to complex human environments—demands systems engineering approaches that can integrate these diverse capabilities into coherent, reliable, robust, and resilient frameworks with performance guarantees

Systems engineering principles provide the ideal framework for implementing the vision of Executive Order 14179 by:

- Accelerating innovation through modular, interoperable components that enable rapid iteration
- Ensuring reliability and robustness through rigorous verification and validation frameworks
- Facilitating market-driven development by establishing clear interfaces between components
- Enabling objective evaluation of system performance independent of ideological considerations
- Creating technical and economic multiplier effects through orchestrated integration

This systems approach naturally removes barriers to innovation while establishing the technical foundations necessary for deploying AI at scale across critical domains including defense, manufacturing, healthcare, energy systems, and education. It embodies the dual imperatives that must guide the AI Action Plan: developing robust engineering practices for AI system development (“Engineering for AI”) and leveraging AI capabilities to transform traditional engineering disciplines (“AI for Engineering”).

3. Leveraging Systems Engineering Principles for Advancing American AI Leadership

Maintaining American leadership in AI requires a bidirectional approach: applying systems engineering principles to advance AI capabilities (“Engineering for AI”) while simultaneously

leveraging AI to transform traditional engineering disciplines (“AI for Engineering”). This dual focus creates a virtuous cycle of innovation that accelerates progress in both domains.

3.1. Systems Integration for Enhanced Performance and Market Competitiveness

The integration of AI components into cohesive systems represents a critical differentiator for maintaining American technological leadership. Systems integration

- creates performance multipliers by orchestrating complementary AI capabilities;
- enables market differentiation through unique combinations of components;
- establishes competitive barriers through system-level optimization that competitors cannot easily replicate;
- accelerates innovation by allowing rapid iteration of individual components within stable system architectures.

Historical precedents from sectors including aerospace, defense, energy systems, supply chains, and telecommunications demonstrate how systems engineering approaches have consistently enabled American technological leadership. The same principles can be applied to AI development to extend U.S. advantages against global competitors who are rapidly advancing component-level capabilities.

Case studies of successful systems integration from both defense and commercial sectors illustrate how orchestrating multiple technologies creates capabilities greater than the sum of individual components. University research plays a crucial role in de-risking advanced integration concepts that can then be commercialized by industry partners.

3.2. Orchestration of AI Components in Complex Systems

Effective orchestration of AI components represents a key technical challenge and opportunity for American leadership. This includes:

- Integration across human operators, classical code, AI algorithms, and cyber-physical systems;
- Development of orchestration frameworks that enable complex workflow automation;
- Establishment of objective performance metrics for comprehensive system evaluation;
- Creation of technical and market multiplier effects through seamless component interaction.

Modern agentic AI systems must coordinate multiple sophisticated capabilities—including hierarchical planning, parametric and symbolic reasoning, extrospective perception, and

theory-of-mind mechanisms—to solve complex problems in human-centric environments. These advanced cognitive capabilities require orchestration frameworks that can maintain coherence across multiple levels of abstraction while ensuring reliable performance in high-consequence applications. American leadership in developing these orchestration frameworks would establish a competitive advantage that transcends individual model capabilities.

Organizations implementing agentic AI systems without proper systems engineering frameworks encounter significant challenges:

- Integration difficulties between multiple AI components;
- Performance bottlenecks that limit overall system capabilities;
- Reliability and robustness issues when deployed in real-world environments;
- Verification and validation challenges for complex, interconnected systems;
- Barriers to scaling from laboratory demonstrations to production environments.

These challenges highlight why systems engineering must serve as the foundation for American AI leadership. Countries and organizations that master systems integration will maintain competitive advantages even as individual AI components become more widely accessible.

3.3. Emerging AI Engineering Patterns as Strategic Frameworks for Advanced Agentic Systems

As AI systems mature toward the sophisticated agentic capabilities we envision, distinct engineering patterns have emerged that provide structured approaches to system development. Six key patterns have particular strategic importance for maintaining American leadership in developing these advanced AI architectures:

- **Models:** Frameworks for developing, evaluating, and deploying foundation models that support both high-fidelity simulations, control and optimization, and symbolic and parametric reasoning capabilities as building blocks for more complex integrated AI systems.
- **Users:** Methodologies for engineering reliable, intuitive interactions between humans and AI systems with theory-of-mind capabilities that can navigate human-centric dynamics while balancing autonomy with appropriate oversight.
- **Context:** Techniques for grounding AI capabilities in relevant domain knowledge through approaches like hierarchical memory structures, advanced knowledge retrieval systems, and fundamental physical, chemical, and biological laws.
- **Orchestration:** Architectures for coordinating multiple cognitive functions including planning, perception, reasoning, goal-setting, control, and optimization that collaborate to solve complex problems through specialized, complementary capabilities.
- **Evals & Observability:** Frameworks for continuous assessment and improvement of multi-level AI systems through rigorous testing and monitoring across different abstraction layers.

- **Mindset:** Organizational practices and engineering culture that maximize effectiveness in developing AI systems with introspective and allocentric reasoning capabilities.

These patterns provide structured approaches to challenges that all advanced AI systems must address. By establishing leadership in these patterns, the United States can create reusable frameworks that accelerate innovation while ensuring reliability across diverse applications. The mindset of modularity inherent in these patterns enables market-driven innovation while maintaining system-level integration capabilities.

3.4. AI Lifecycle Management (AIOps) as Competitive Advantages

The operational infrastructure for developing, deploying, and maintaining AI systems represents a critical enabler for rapid innovation:

- Extend MLOps concepts to an AIOps framework that streamline the transition from research to production through standardized processes;
- Lifecycle management ensures consistent performance through continuous monitoring and improvement, self-healing, self-tuning, and adaptation to new environments;
- Model governance frameworks maintain technical quality and reliability throughout system development and deployment
- Integration of DevSecOps principles ensures security and reliability in national security applications

Current global competition in AI is not merely about model capabilities but increasingly about the infrastructure that enables rapid iteration, deployment, and continuous improvement and adaptation. U.S. leadership in AIOps frameworks would create a sustainable competitive advantage in AI system development and deployment velocity.

4. Strategic Recommendations for the AI Action Plan

4.1. Research Priorities that Enable Market-Driven AI Leadership

To maintain American AI leadership, we recommend prioritizing research investments in both “Engineering for AI” and “AI for Engineering”:

4.1.1 Engineering for AI:

- Systems-level integration frameworks that accelerate the composition of advanced agentic capabilities including hierarchical planning, symbolic reasoning, control, optimiza-

- tion, and theory-of-mind mechanisms;
- Verification, validation, and testing methodologies for complex AI systems with multiple cognitive functions operating across different levels of abstraction;
- Interoperability standards that enable competition while facilitating the integration of diverse AI components including perception, reasoning, and planning systems;
- Performance guarantees for agentic AI systems operating in safety-critical domains with human-centric dynamics;
- Cognitive architectures that support allocentric reasoning and adaptive goal-setting in complex environments;
- Energy-efficient computing paradigms for supporting sophisticated AI cognition at scale.

4.1.2 AI for Engineering:

Establishing U.S. leadership in AI will necessitate an integration of AI tools in every engineering disciplines from supply chains and manufacturing, to energy production and transmission, healthcare, material design, semiconductor design and implementation. This will require

- AI-driven tools that accelerate traditional engineering processes across disciplines, by bringing orders of magnitude improvements in efficiency.
- Frameworks for integrating AI capabilities into existing engineering workflows, integrating domain-inspired AI systems.
- AI systems that enhance human engineering capabilities rather than replacing them, including in design, monitoring, and optimization.
- Methodologies for addressing engineering challenges that were previously intractable, through the combination of data- and model-driven approaches.
- Simulations and digital twins, and their surrogate models, that enable rapid exploration of design spaces

Universities represent ideal and unique environments for high-risk, exploratory research needed to develop novel integration approaches and foundational systems engineering frameworks. By focusing federal investments on these system-level capabilities, the AI Action Plan can establish enduring American advantages that are difficult for competitors to replicate.

We recommend establishing a national research initiative focused specifically on systems engineering for agentic AI, with coordinated funding across NSF, DARPA, DOE, and other federal agencies. This initiative should emphasize market-driven approaches that preserve competition while establishing frameworks that accelerate commercialization.

4.2. Workforce Development and Educational Transformation

Maintaining AI leadership requires addressing the complementary challenges of developing "Workforce-ready AI" systems designed to integrate seamlessly with human workers and creating an "AI-ready Workforce" capable of effectively implementing and working alongside advanced AI systems:

- **Workforce-ready AI:** Developing AI systems that augment human capabilities, align with existing workflows, and create intuitive interfaces for non-technical users - enabling workers to achieve greater productivity and innovation that strengthens America's competitive edge.
- **AI-ready Workforce:** Building human capacity through educational programs that prepare workers at all levels to effectively implement, work alongside, and maximize the potential of AI systems - creating economic opportunities while enhancing national technological capabilities.

Addressing these dual challenges requires strategic approaches to workforce development:

4.2.1 Paradigm Shift in Education for AI Systems Engineering

Developing an AI-ready workforce requires a fundamental reconceptualization of technical education from single-degree attainment to lifelong learning cycles. Georgia Tech Professional Education (GTPE) provides a model for continuous upskilling aligned with rapidly evolving AI capabilities.

This educational transformation must address a number of key paradigm shifts:

- From training narrow AI tool specialists to developing systems specialists capable of orchestrating complex AI-enabled workflows.
- From siloed technical skills to cross-domain mastery integrating computer science, cognitive science, engineering, and ethics.
- From theoretical foundations alone to balanced curricula that integrate practice, implementation, and real-world problem-solving.
- From isolated technical skills to interdisciplinary understanding that bridges domains like computer science, cognitive science, engineering, and ethics.
- From static curricula to adaptive learning pathways that maintain American competitive advantage amid rapid AI advancement
- From traditional assessment methods to competency-based evaluation focused on system integration capabilities.

Georgia Tech's AI education initiatives incorporate experiential learning environments that simulate the complex challenges of real-world AI system integration. These environments

provide students with hands-on experience orchestrating multiple AI components to solve problems across domains from healthcare to manufacturing to defense applications. Georgia Tech is also a national leader in high-school AI education and the training faculty across the country to create novel AI programs. Both components are key to the future of AI in the U.S., attracting talent early on and training the educators needed to sustain the workforce demands.

4.2.2 Developing Workforce-Ready AI Systems

The complementary challenge to building an AI-ready workforce is creating AI systems that are designed from the ground up to integrate effectively with human workers. Workforce-ready AI systems should:

- Provide intuitive interfaces that empower users to easily adopt and leverage AI capabilities;
- Augment human capabilities to amplify productivity, creativity, and innovation, rather than simply automating existing processes;
- Seamlessly integrate with and adapt to organizational workflows to enhance efficiency and effectiveness
- Offer appropriate explanations of AI-driven decisions to build trust with human operators;
- Enable progressive levels of autonomy that evolve as organizational readiness increases.

Georgia Tech’s research in human-AI collaboration and interactive machine learning provides frameworks for developing AI systems that work effectively alongside humans with varying technical backgrounds and domain expertise.

We recommend establishing a national program that incentivizes the development of workforce-ready AI systems through targeted research funding, public-private partnerships, and educational initiatives that bridge technical and domain-specific training.

4.3. Computing Infrastructure and Technical Standards

Next-generation AI capabilities require corresponding advances in computing infrastructure for commercial and national security resilience:

- Low-power and low-latency computing innovations enabling AI at the edge;
- Distributed computing networks that manage dynamic mission-critical resource allocation, from commercial robotics to tactical edge environments;
- Energy-efficient architectures that create both economic and strategic advantages;

- Technical standards for interoperability that enable component-level competition;
- Secure computing environments for high-consequence AI applications.

University-industry partnerships play a crucial role in accelerating commercialization of computing innovations that enable advanced AI capabilities. Technical standards should focus on interfaces and interoperability rather than implementation details to preserve market dynamics.

We recommend investments in next-generation computing infrastructure specifically optimized for agentic AI systems, with particular emphasis on:

- Energy-efficient architectures that reduce computational overhead;
- Edge computing capabilities that enable real-time AI processing;
- Communications infrastructure that facilitates distributed AI systems;
- Standards frameworks that ensure interoperability across vendors;
- hardware architectures and accelerators that support the integration of data-driven AI with reasoning, optimization, and scientific computing.

4.4. AI for Scientific Discovery and Workflow Transformation

AI systems are fundamentally transforming scientific discovery and engineering workflows across disciplines by enabling data-driven approaches that complement traditional theoretical and experimental methods:

- Systems engineering principles provide frameworks for reimagining scientific and engineering workflows to leverage AI capabilities (“AI for Engineering”);
- AI-driven scientific discovery pipelines accelerate innovation in fields from materials science to drug development;
- Enhanced capabilities for abstraction and modeling of complex phenomena through AI-augmented approaches transform conventional engineering disciplines;
- Integration of domain-specific scientific tools with general-purpose AI capabilities creates powerful new research methodologies;
- Traditional engineering fields like energy systems and supply chain engineering benefit particularly from data structure-aware AI applications.

The economic multiplier effects of accelerated scientific discovery represent a critical dimension of national competitiveness. By establishing leadership in AI-driven scientific workflows, the United States can accelerate breakthrough innovations across sectors ranging from advanced materials to pharmaceutical development to climate science. Systems engineering approaches are essential for integrating AI capabilities into scientific workflows while preserving methodological rigor and ensuring reproducibility.

5. Implementation Roadmap for 180-Day AI Action Plan

We recommend that the 180-day AI Action Plan include the following concrete actions:

1. Establish a National AI Systems Engineering Initiative

- Create a cross-agency coordination mechanism for systems engineering research;
- Develop roadmaps for key technical capabilities required for agentic AI systems;
- Establish metrics and benchmarks for evaluating systems-level AI performance.

2. Launch Workforce and Talent Development Programs

- Fund educational initiatives focused on developing an AI Systems Engineering talent pool;
- Create frameworks for AI systems education across educational levels;
- Establish certification programs for AI systems engineering professionals.

3. Create Technical Standards Frameworks

- Develop interoperability standards for AI systems and traditional software components;
- Establish performance metrics for AI system evaluation;
- Create verification and validation frameworks for agentic AI systems.

4. Facilitate Technology Transfer Mechanisms

- Streamline pathways from university research to commercial deployment
- Establish public-private partnerships for AI systems development, training, and deployment;
- Create incentive structures for the commercial adoption of AI systems.

5. Implement Computing Infrastructure Investments

- Fund next-generation computing architectures for AI systems
- Develop energy-efficient computing platforms for AI applications
- Create distributed computing frameworks for collaborative AI development

Success should be measured across multiple dimensions, including technical capabilities, economic competitiveness, and national security enhancements. Regular assessment against these metrics will ensure that the AI Action Plan delivers on its objectives to enhance American AI leadership.

6. Conclusion

A comprehensive AI Action Plan must champion both “AI for Engineering” and “Engineering for AI”—recognizing that maintaining American leadership requires not only advancing AI capabilities through robust engineering practices but also leveraging AI to transform

traditional engineering disciplines. This bidirectional approach creates a virtuous cycle of innovation that accelerates progress across domains.

The emergence of sophisticated agentic AI systems—with advanced planning capabilities, hierarchical memory, symbolic and parametric reasoning mechanisms, and multi-faceted theories-of-mind—represents both the most significant opportunity and the most profound challenge for American technological leadership. These systems will transcend today’s language models to become adaptable cognitive architectures capable of navigating complex human-centric environments in high-consequence domains. By embracing systems engineering principles as the foundation for advancing these sophisticated AI capabilities, the United States can establish sustainable competitive advantages in developing agentic systems that perform increasingly autonomous functions while maintaining reliability, robustness, and alignment with human objectives.

Georgia Tech stands ready to support the development and implementation of this plan through our research capabilities, educational programs, and industry partnerships. By adopting systems engineering principles as the foundation for AI advancement, the United States can establish sustainable competitive advantages that fulfill the vision of Executive Order 14179 to “sustain and enhance America’s global AI dominance.”